Question 8: Downloading and installing Hive; startup scripts

Aim

To install Apache Hive, configure its environment and start the Hive services for executing HiveQL queries.

Procedure

1. Install and configure Hadoop first, and ensure HDFS and YARN are working.

2. Install a Java version supported by the chosen Hive release. Download the Apache Hive binary distribution and

extract it, for example, to /usr/local/hive.

3. Set HIVE_HOME and add its bin directory to PATH.

4. Configure hive-site.xml with the metastore connection and warehouse directory. For a lab installation, Derby can be

used as an embedded metastore; a production setup should use a supported external database.

5. Create the warehouse directory in HDFS and grant suitable permissions.

6. Initialize the metastore schema once, then start Hive using the Hive CLI or Beeline with the appropriate service.

7. Run a simple SHOW DATABASES query to verify the installation.

Environment and configuration

```
# ~/.bashrc

export HIVE_HOME=/usr/local/hive

export PATH=$PATH:$HIVE_HOME/bin

export HADOOP_HOME=/usr/local/hadoop

<!-- $HIVE_HOME/conf/hive-site.xml (illustrative embedded Derby setup) -->

<configuration>

<property>

<name>javax.jdo.option.ConnectionURL</name>

<value>jdbc:derby:;databaseName=metastore_db;create=true</value>

</property>

<property>

<name>hive.metastore.warehouse.dir</name>

<value>/user/hive/warehouse</value>

</property>
```

</configuration>

Startup commands

source ~/.bashrc

hdfs dfs -mkdir -p /user/hive/warehouse

hdfs dfs -chmod 777 /user/hive/warehouse

schematool -dbType derby -initSchema # First-time setup only

hive

At the hive prompt:

SHOW DATABASES;

Or use Beeline when HiveServer2 is configured:

beeline -u jdbc:hive2://localhost:10000/default

Output of Program

hive> SHOW DATABASES;

OK

default

Time taken: 1.2 seconds

Result

Hive is installed and initialized. The Hive prompt opens and executes HiveQL commands successfully.

Version note: Hive, Hadoop and Java compatibility varies by release. Use a mutually compatible combination specified by the

laboratory; do not assume the illustrative paths or Java version apply to every release.